DR.BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

DEPARTMENT OF ELECTRICAL ENGINEERING

SEMISTER EXAMINATION 2019

Subject- Electrical Utilization (EE801) **Semester: VIII**
Date: 14-05-2019 **Time: 3Hour** **Marks: 70**

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Instructions:
1) Question no 1 is compulsory.
2) From remaining question Q.2 to Q.7 solve any five.
3) If require make appropriate assumptions with clear explanation.
4) Solve the questions in sequential manner.

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Q.1 **Answer the following question. (Any Two)** **(10)**

A) Explain Laws of illumination (5)

B) Explain faraday's laws of electrolysis? (5)

C) Write a short note on Electroplating process? (5)

Q.2) **Answer the following question. (Any Two)** **(12)**

A) Explain characteristic of different types of mechanical loads?

B) Describe load equalization and derive its expression?

C) A three phase 50Hz,4 pole ,20 Hz induction motor has a uniform load torque of 15kg-m and extra load torque of 80 kg-m for 8 sec motor full load slip 2% and efficiency 75% .Calculate

i) Weight of flywheel, if the radius of gyration is 0.8, motor torque not to increase more than twice the full load torque.

ii) Horse power output at maximum load.

Q.3) **Answer the following question.**

A) If 96500 C of electricity liberates 1gm. Equivalent of any substance, how long it will take for a current of 0.15A to deposit 20mgm of copper form a solution of copper sulphate. (6)

Chemical equivalent of copper is to be taken as 32

B) Explain factor govering electro deposition processes? (6)

Q.4) **Answer the following question.** (12)

A) Define following terms

i) Luminous Intensity

ii) Coefficient of Utilization

iii) Depreciation Factor

iv) Radiant Efficiency

B) Two lamps each of 500W with a lamp efficiency of 25 lumens per watt area mounted on two lamp posts 10m apart. The posts having different height of 3m and 4m respectively. Calculate the illumination at post mid-way between the lamp posts.

Q.5) **Answer the following question. (Any Two)** (12)

A) Explain ajax wyatt furnace in detail?

B) Classify different types of welding and explain metal inert gas (MIG) welding in detail?

C) Explain seam welding in detail also draw the equipment used in arc welding?

Q.6) **Answer the following question.** (12)

A) Explain speed time curve for urban service traction? (6)

B) Explain current collectors for over -head contact wire in traction? (6)

Q.7) **Solve the following questions (Any Two)** (12)

A) Explain different types of motor used traction with their characteristic?

B) Compare AC and DC Traction system?

C) Write a short note on regenerative breaking used in traction system?